Incorporating Visual Non-Detection Information in Cooperative Localization of an ROV in a Low-Visibility Lake Environment

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***Abstract*— This paper presents an approach to localizing a Remotely Operated Vehicle (ROV) paired with an Unmanned Surface Vehicle (USV) using a camera onboard the ROV in low-visibility environments. A combination of Extended Kalman Filter (EKF) and grid-based estimators are used to allow the non-Gaussian, negative information provided when the ROV is not able to detect the USV to be incorporated into the estimate of the ROV's location. The only other sensor used to estimate the ROV's position is an Inertial Navigation System (INS). The technique was validated with a pair of experiments run in a lake on a vehicle team consisting of a BlueROV and a custom built USV.**

I. Introduction

ROVs are an invaluable tool for collecting data and performing inspections in a variety of applications from environmental monitoring to mapping and vehicle and infrastructure inspections []. To be able to perform these operations, and especially to enable future autonomous implementations of them, it is essential for the ROV to be able to accurately determine its position. This is especially challenging in environments such as lakes or piers where Global Positioning System (GPS) cannot be used due to the attenuation of radio frequencies through the water, sonar based positioning is limited by reflections off of the surface and floor (as well as the availability of sonars to begin with), and visual systems are limited by low visibility through turbid waters.

There are several strategies to localizing an ROV visually. [1], [2] use Simultaneous Localization And Mapping (SLAM) to generate a map of the environment the ROV or ROV is working in and to determine its position against that map. This approach is highly applicable when there is sufficient visibility or the vehicle remains close to fixed features, however, when it moves to somewhere without fixed references, it is left to fall back on dead reckoning approaches. Visual odometry is used in [3] with stereo cameras to estimate the vehicle's motion. This suffers the same issue as SLAM when there are not fixed references close enough to observe, without the ability of SLAM to recover the position when a reference does become available. That said, visual odometry has the benefit of not requiring the construction and maintenance of an entire map of the environment. Cooperative localization using visual relative positioning is demonstrated in [4], [5]. This approach both allows all vehicles to reduce their position uncertainty by fusing information from more sources, it also allows a vehicle with a better known position to be used as a reference for other vehicles. This latter feature allows an underwater vehicle to get fixed references from a surface vehicle that it can see an which has a GPS, allowing it to get absolute position references without the need for a map. A key drawback is the need for the vehicles to be able to communicate, which, while normally challenging underwater, with an ROV, is facilitate readily by the tether. Finally, in [6] all of the above strategies are combined.

All of these approaches make use of positive detections from the camera, whether a detection of the environment for SLAM and odometry or of another vehicle for cooperative localization. However, there is additional information to be gained about the vehicle's location when the camera does not detect a known reference. Given the low-visibility in many underwater environments, and especially near- and on-shore environments like lakes, rivers, and piers, the non-detection case becomes frequent, so retaining the information it provides is essential for localization.

This work aims to demonstrate the incorporation of this non-detection information from a cooperative localization approach in the localization of an ROV operating in a low-visibility lake environment. Figure 1 shows the ROV that is to be localized and the supporting USV.

Fig. 1. ROV tethered to a support USV. The ROV is a BlueRobotics BlueROV2 and the USV is custom built.

The next section lays out the cooperative localization framework that allows the non-detection information to be incorporated into the localization of the ROV. Section IV describes experiments performed in a lake to demonstrate the technique and the results of these experiments. Finally,

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section V concludes with ongoing work.

II. ROV State Estimation

A. Problem Formulation

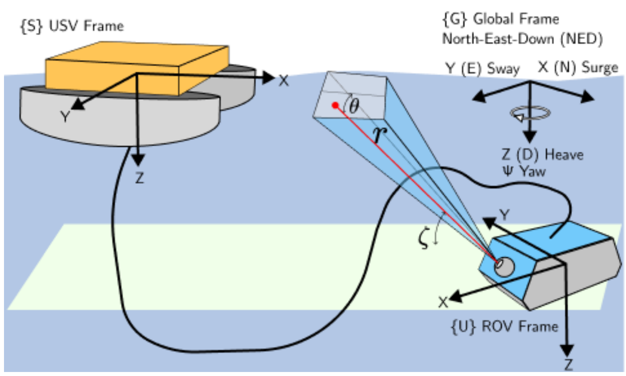


Fig. 2. Global and vehicle coordinate frames. The camera on the ROV is able to observe the USV to get a relative position when the USV is within its FOV.

Figure 2 shows a schematic diagram of the ROV-USV system together with the coordinate frames used to localize the individual vehicles. The global frame (denoted ${}^{\{G\}}(\cdot)$) is aligned with x, y, z along North, East, Down and located at a designated reference point. The vehicle frames (${}^{\{S\}}(\cdot)$ for the USV and ${}^{\{U\}}(\cdot)$ for the ROV) are similarly aligned with x along the longitudinal axis of the vehicle, y facing to the right, and z down, with the origin located at the center of mass of the vehicle. r, ζ, and θ represent the range, azimuth, and bearing respectively to any point within the field of view (FOV) of the camera where it projects onto the horizontal plane.

The goal of this system is to localize the ROV reliably through cooperation. The localization problem can be mathematically stated using the notations and formulations first introduced in [7]. Since the ROV is stably oriented in a horizontal plane, and the USV is on the water surface (which is assumed to be calm, and therefore flat), the state variables of each vehicle to estimate may be given by

$$
\begin{aligned}
{}^{\{G\}}\mathbf{x}_k^u &\equiv \begin{bmatrix} x^u & y^u & z^u & \psi^u & \dot{x}^u & \dot{y}^u \end{bmatrix}^\top \\
{}^{\{G\}}\mathbf{x}_k^s &\equiv \begin{bmatrix} x^s & y^s & \psi^s \end{bmatrix}^\top,
\end{aligned}
$$

where ψ is the yaw angle of the vehicle relative to North. It is to be noted that $\{G\}$ is dropped subsequently in the paper to avoid notational complexity.

The discrete motion of vehicle $r \in \{u, s\}$, given control input $\mathbf{u}_k^r$ and motion noise $\mathbf{w}_k^r$, is probabilistically described:

$$
\dot{\mathbf{x}}_k^r = \mathbf{f}^r\left(\mathbf{x}_k^r, \mathbf{u}_k^r, \mathbf{w}_k^r\right) \tag{1a}
$$

$$
\mathbf{x}_k^r \approx \mathbf{x}_{k-1}^r + \dot{\mathbf{x}}_{k-1}^r \Delta t, \tag{1b}
$$

where Δt is a small time increment. The vehicle r is equipped with any effective global sensors such as GPS and INS, which could localize the vehicle itself. The ith sensor is generically described as:

$$
{}^{r_i}\mathbf{z}_k^r = {}^{r_i}\mathbf{h}^r\left(\mathbf{x}_k^r, {}^{r_i}\mathbf{v}_k^r\right), \tag{2}
$$

where ${}^{r_i}\mathbf{z}_k^r$ and ${}^{r_i}\mathbf{v}_k^r$ are the observation and the sensor noise respectively. Since the two vehicles have a wired connection, all the observations are shared between them.

The vehicle r can also be localized when the other vehicle $q (\neq r)$ detects the vehicle r using their local sensors, each with an "observable region", also known as an FOV. Defining the probability of detecting the vehicle r $P_d\left(\mathbf{x}_k^r | \mathbf{x}_k^q; \boldsymbol{\pi}^{q_j}\right)$ with the parameters associated with the sensor capability $\boldsymbol{\pi}^{q_j}$, the observable region of the jth sensor can be expressed as

$$
{}^{q_j}\mathcal{X}_o^r = \left\{\mathbf{x}_k^r | 0 < P_d\left(\mathbf{x}_k^r | \mathbf{x}_k^q; \boldsymbol{\pi}^{q_j}\right) \leq 1\right\}. \tag{3}
$$

Accordingly, the observation from the jth sensor is given by

$$
{}^{q_j}\mathbf{z}_k^r = \begin{cases} {}^{q_j}\mathbf{h}^r\left(\mathbf{x}_k^r, \mathbf{x}_k^q, {}^{q_j}\mathbf{v}_k^r\right) & \mathbf{x}_k^r \in {}^{q_j}\mathcal{X}_o^r \\ \varnothing & \text{otherwise} \end{cases}, \tag{4}
$$

where $\varnothing$ represents an "empty element," indicating that the other vehicle is unobservable.

B. Recursive Bayesian Estimation (RBE) Techniques

RBE updates and maintains the belief of the state of concern by recursively performing prediction and correction. Let the initial belief of the robot's state be represented with a probability density function as $p(\mathbf{x}_0^r)$. Given a series of observations up to time k from the self-localization sensors, ${}^{r}\tilde{\mathbf{z}}_{1:k-1}^r \equiv \left\{{}^{r_i}\tilde{\mathbf{z}}_{1:k-1}^r | \forall i \in \mathcal{I}^r\right\}$ and those from the other vehicle ${}^{q}\tilde{\mathbf{z}}_{1:k-1}^r \equiv \left\{{}^{q_j}\tilde{\mathbf{z}}_{1:k-1}^r | \forall j \in \mathcal{I}^q\right\}$ in addition to the last belief of the robot's state $p(\mathbf{x}_{k-1}^r | {}^{r_i}\tilde{\mathbf{z}}_{1:k-1}^r, {}^{q_j}\tilde{\mathbf{z}}_{1:k-1}^r)$, the prediction step estimates the belief at time k using the Chapman-Kolmogorov equation:

$$
\begin{aligned}
&p(\mathbf{x}_k^r | {}^{r}\tilde{\mathbf{z}}_{1:k-1}^r, {}^{q}\tilde{\mathbf{z}}_{1:k-1}^r) = \\
&\qquad \int p(\mathbf{x}_k^r | \mathbf{x}_{k-1}^r) p(\mathbf{x}_{k-1}^r | {}^{r}\tilde{\mathbf{z}}_{1:k-1}^r, {}^{q}\tilde{\mathbf{z}}_{1:k-1}^r) d\mathbf{x}_{k-1}^r,
\end{aligned} \tag{5}
$$

where $p(\mathbf{x}_k^r | \mathbf{x}_{k-1}^r)$ is the transition probability resulting from the motion model in Eq. (1b).

The correction step incorporates new observations to improve the estimate of the vehicle's state:

$$
\begin{aligned}
&p(\mathbf{x}_k^r | {}^{r}\tilde{\mathbf{z}}_{1:k}^r, {}^{q}\tilde{\mathbf{z}}_{1:k}^r) \propto \\
&\qquad p(\mathbf{x}_k^r | {}^{r}\tilde{\mathbf{z}}_{1:k-1}^r, {}^{q}\tilde{\mathbf{z}}_{1:k-1}^r) l(\mathbf{x}_k^r | {}^{r}\tilde{\mathbf{z}}_k^r, {}^{q}\tilde{\mathbf{z}}_k^r),
\end{aligned} \tag{6}
$$

where $l(\mathbf{x}_k^r | {}^{r}\tilde{\mathbf{z}}_k^r, {}^{q}\tilde{\mathbf{z}}_k^r)$ is the joint likelihood of the observations of the vehicle's state, which is given by

$$
l(\mathbf{x}_k^r | {}^{r}\tilde{\mathbf{z}}_k^r, {}^{q}\tilde{\mathbf{z}}_k^r) = \prod_{i \in \mathcal{I}^r} l(\mathbf{x}_k^r | {}^{r_i}\tilde{\mathbf{z}}_k^r) \prod_{j \in \mathcal{I}^q} l(\mathbf{x}_k^r | {}^{q_j}\tilde{\mathbf{z}}_k^r). \tag{7}
$$

The individual likelihoods come from the sensor models (2) and (4).

1) Grid-Based Method: The grid-based method effectively processes non-Gaussian beliefs by representing the belief space by a set of grid cells. For each cell, an associated probability density is assigned such that they collectively approximate a belief:

$$p(\mathbf{x}) \approx \sum_i p(\hat{\mathbf{x}}_i^r)\delta_i(\mathbf{x}-\hat{\mathbf{x}}_i^r), \tag{8}$$

where $\hat{\mathbf{x}}_i^r$ is the center of the ith grid cell and $\delta_i(\cdot)$ is the Dirac delta function. With the approximation, the prediction (5) can be performed efficiently through convolution, though the grid-based method is generally expensive in computation:

$$\begin{aligned} &p(\hat{\mathbf{x}}_{i,k}^r | {}^r\tilde{\mathbf{z}}_{1:k-1}^r, {}^q\tilde{\mathbf{z}}_{1:k-1}^r) = \\ &\qquad p(\hat{\mathbf{x}}_{j,k}^r | \hat{\mathbf{x}}_{i,k-1}^r) * p(\hat{\mathbf{x}}_{i,k-1}^r | {}^r\tilde{\mathbf{z}}_{1:k-1}^r, {}^q\tilde{\mathbf{z}}_{1:k-1}^r), \end{aligned} \tag{9}$$

where $*$ is the convolution operator and $p(\hat{\mathbf{x}}_{j,k}^r | \hat{\mathbf{x}}_{j,k-1}^r)$ is the jth element of the convolution matrix.

The correction of the grid-based method is straightforward, which processes Eq. (6) for each cell:

$$\begin{aligned} &p(\hat{\mathbf{x}}_{i,k}^r | {}^r\tilde{\mathbf{z}}_{1:k}^r, {}^q\tilde{\mathbf{z}}_{1:k}^r) \propto \\ &\qquad p(\hat{\mathbf{x}}_{i,k}^r | {}^r\tilde{\mathbf{z}}_{1:k-1}^r, {}^q\tilde{\mathbf{z}}_{1:k-1}^r) l(\hat{\mathbf{x}}_{i,k}^r | {}^r\tilde{\mathbf{z}}_k^r, {}^q\tilde{\mathbf{z}}_k^r). \end{aligned} \tag{10}$$

Note that the observation likelihood and the updated belief become heavily non-Gaussian when non-detection events are handled positively; the observation likelihood from the sensor model (4) is given by:

$$l(\mathbf{x}_k^r | {}^{q_j}\tilde{\mathbf{z}}_{1:k}^r) = \begin{cases} p\left(\mathbf{x}_k^r | {}^{q_j}\tilde{\mathbf{z}}_{1:k}^r, \mathbf{x}_k^q; \boldsymbol{\pi}^{q_j}\right) & {}^{q_j}\tilde{\mathbf{z}}_{1:k}^r \in {}^{q_j}\mathcal{X}_d^r \\ 1 - P_d(\mathbf{x}_k^t | \mathbf{x}_k^{q_j}; \boldsymbol{\pi}^{q_j}) & \text{otherwise}, \end{cases} \tag{11}$$

where ${}^{q_j}\mathcal{X}_d^r = \left\{\mathbf{x}_k^t | {}^{q_j}\epsilon^r < P_d(\mathbf{x}_k^r | \mathbf{x}_k^{q_j}, \boldsymbol{\pi}^{q_j}) \leq 1\right\}$ is the detectable region, which is a subset of the observable region narrowed down with the level of confidence ${}^{q_j}\epsilon^r$ [8]. The first case with detection may be approximated by a Gaussian distribution, but the second case with no detection makes the observation likelihood heavily non-Gaussian.

2) Extended Kalman Filter: EKF handles non-linear motion and sensor models, but requires the assumption of Gaussian noise, causing Eqs. (1a), (2), (4) to be reformulated:

$$\dot{\mathbf{x}}_k^r = \mathbf{f}^r\left(\mathbf{x}_k^r, \mathbf{u}_k^r\right) + \mathbf{w}_k^r \tag{12a}$$

$${}^{r_i}\mathbf{z}_k^r = {}^{r_i}\mathbf{h}^r\left(\mathbf{x}_k^r\right) + {}^{r_i}\mathbf{v}_k^r \tag{12b}$$

$${}^{q_j}\mathbf{z}_k^r = {}^{q_j}\mathbf{h}^r\left(\mathbf{x}_k^r, \mathbf{x}_k^q\right) + {}^{q_j}\mathbf{v}_k^r, \tag{12c}$$

where $\mathbf{w}_k^r \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_k^{rw})$, ${}^{r_i}\mathbf{v}_k^r \sim \mathcal{N}(\mathbf{0}, {}^{r_i}\boldsymbol{\Sigma}_k^{rv})$, and ${}^{q_j}\mathbf{v}_k^r \sim \mathcal{N}(\mathbf{0}, {}^{q_j}\boldsymbol{\Sigma}_k^{rv})$. The prediction updates the mean and the covariance of the belief in time as

$$\bar{\mathbf{x}}_{k|k-1}^r = \mathbf{f}^r\left(\bar{\mathbf{x}}_{k-1|k-1}^r, \tilde{\mathbf{u}}_k^r\right) \tag{13a}$$

$$\boldsymbol{\Sigma}_{k|k-1}^r = \nabla_r\mathbf{f}^r\boldsymbol{\Sigma}_{k-1|k-1}^r\nabla_r\mathbf{f}^{r\top} + \boldsymbol{\Sigma}_k^{rw}, \tag{13b}$$

where $\nabla_r\mathbf{f}^r$ is the Jacobian of $\mathbf{f}^r(\cdot)$ with respect to the vehicle evaluated at the estimate $\bar{\mathbf{x}}_{k-1|k-1}^r$. The correction updates the mean and the covariance of the belief in observation as

$$\begin{aligned} \bar{\mathbf{x}}_{k|k}^r = \bar{\mathbf{x}}_{k|k-1}^r &+ \sum_i {}^{r_i}\mathbf{K}_k^r\left[{}^{r_i}\tilde{\mathbf{z}}_k^r - {}^{r_i}\mathbf{h}^r\left(\hat{\mathbf{x}}_{k|k-1}^r\right)\right] \\ &+ \sum_j {}^{q_j}\mathbf{K}_k^r\left[{}^{q_j}\tilde{\mathbf{z}}_k^r - {}^{q_j}\mathbf{h}^r\left(\hat{\mathbf{x}}_{k|k-1}^r\right)\right] \end{aligned} \tag{14a}$$

$$\boldsymbol{\Sigma}_{k|k}^r = \left(\mathbf{I} - \textstyle\sum_i {}^{r_i}\mathbf{K}_k^r\nabla_r{}^{r_i}\mathbf{h}^r - \sum_j {}^{q_j}\mathbf{K}_k^r\nabla_r{}^{q_j}\mathbf{h}^r\right)\boldsymbol{\Sigma}_{k|k-1}^r, \tag{14b}$$

where ${}^{(\cdot)}\mathbf{K}_k^r$ is the Kalman gain given by

$${}^{(\cdot)}\mathbf{K}_k^r = \nabla_r{}^{(\cdot)}\mathbf{h}^{r\top}\left(\nabla_r{}^{(\cdot)}\mathbf{h}^r\boldsymbol{\Sigma}_{k|k-1}^r\nabla_r{}^{(\cdot)}\mathbf{h}^{r\top} + {}^{(\cdot)}\boldsymbol{\Sigma}_k^{rv}\right)^{-1}. \tag{15}$$

While computationally efficient, the problem of the EKF is the Gaussian assumption, which is not valid for cases including non-detection events.

III. ROV Localization Using Non-Detections from Camera

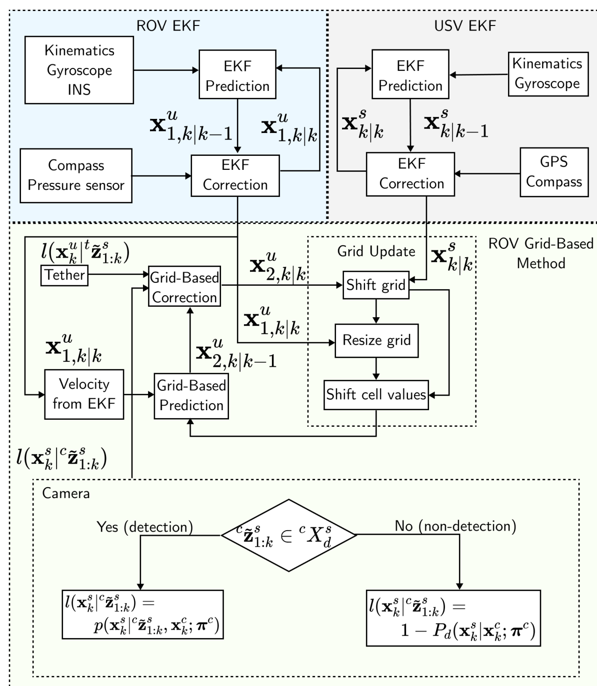


Fig. 3. Combination of EKF and grid-based method to incorporate non-detection information from the ROV's camera.

Figure 3 shows how the estimation is performed to make use of the non-detection events from a camera mounted on the ROV while maintaining computational speed. The states of the ROV are separated into two different sets to be estimated by different methods, EKF for the states that can be observed with only Gaussian observations and grid-based method for the states that can not be (namely, the position in the horizontal plane of the ROV):

TABLE I

SENSORS FOR LOCALIZATION

Vehicle	Sensor	Data	State to est.	Data
ROV	Pressure	z_k^u	Ver. pos.	z_k^u
	Compass	ψ_k^u	Heading	ψ_k^u
	INS	$[\ddot{x}_k^s, \ddot{y}_k^s]$	Hor. acc.	$[\ddot{x}_k^s, \ddot{y}_k^s]$
	Camera	$\left[{}^{\{U\}}x_k^s, {}^{\{U\}}y_k^s\right]$	Hor. pos.	$\left[{}^{\{U\}}x_k^s, {}^{\{U\}}y_k^s\right]$ $\left[{}^{\{S\}}x_k^u, {}^{\{S\}}y_k^u\right]$
USV	Compass	ψ_k^s	Heading	ψ_k^s
	GPS	$[x_k^s, y_k^s]$	Hor. pos.	$[x_k^s, y_k^s]$

$$\mathbf{x}_k^u = \begin{bmatrix}\mathbf{x}_{1,k}^{u}{}^\top & \mathbf{x}_{2,k}^{u}{}^\top\end{bmatrix}^\top$$
$$\mathbf{x}_{1,k}^u = \begin{bmatrix}z_k^u & \psi_k^u & \dot{x}^u & \dot{y}^u\end{bmatrix}^\top$$
$$\mathbf{x}_{2,k}^u = \begin{bmatrix}x_k^u & y_k^u\end{bmatrix}^\top,$$

with associated motion models:

$$\dot{\mathbf{x}_k^u} = \begin{bmatrix}\dot{\mathbf{x}_{1,k}^u}^\top & \dot{\mathbf{x}_{2,k}^u}^\top\end{bmatrix}^\top$$
$$\dot{\mathbf{x}_{1,k}^u} = \begin{bmatrix}\dot{z}_k^u & \dot{\psi}_k^u & \ddot{x}^u & \ddot{y}^u\end{bmatrix}^\top$$
$$\dot{\mathbf{x}_{2,k}^u} = \begin{bmatrix}\dot{x}_k^u & \dot{y}_k^u\end{bmatrix}^\top,$$

where $\mathbf{x}_{1,k}^u$ is fully observable with global sensors, and $\mathbf{x}_{2,k}^u$ may not be observed by local sensors. The state variables of the USV are, on the other hand, handled collectively as $\mathbf{x}_k^s$ due to their full observability with global sensors. The corresponding observations are:

$$\mathbf{z}_k^u = \begin{bmatrix}\mathbf{z}_{1,k}^{u}{}^\top & \mathbf{z}_{2,k}^{u}{}^\top\end{bmatrix}^\top.$$

Note that there is no superscript proceeding the observation since the location of each sensor is trivial.

Table I lists the sensors used to estimate the positions of the two vehicles. The GPS locates the horizontal position of the USV, whereas the ROV position is not directly observable. However, since it identifies the USV position at the surface relative to the frame of the ROV, the camera can also be used to identify the ROV position relative to the USV frame. The position of the ROV, as a result, is estimated using the camera on the ROV, which is a local sensor with limited observable region. The vertical position and the heading of the ROV are estimated using global sensors.

A. EKF of $\mathbf{x}_{1,k|k}^u$ and $\mathbf{x}_{k|k}^s$

First, all of the states of the USV and the depth, heading, and velocity of the ROV are estimated using an EKF. Eqs. (16) and (17) are substituted into Eq. (12a) to perform the prediction step.

$$\begin{bmatrix}\dot{z}_k^u \\ \dot{\psi}_k^u \\ \ddot{x}_k^u \\ \ddot{y}_k^u\end{bmatrix} = \begin{bmatrix}\frac{1}{\Delta t}\left(z_k^u - z_{k-1}^u\right) \\ {}^{\{U\}}\dot{\psi}_k^u \\ {}^{\{G\}}_{\{U\}}\mathbf{R}\,{}^{\{U\}}\ddot{x}_k^u \\ {}^{\{G\}}_{\{U\}}\mathbf{R}\,{}^{\{U\}}\ddot{y}_k^u\end{bmatrix} + \begin{bmatrix}w_z^u \\ {}^{g}v^u \\ {}^{ins}v_x^u \\ {}^{ins}v_y^u\end{bmatrix} \tag{16}$$

$$\begin{bmatrix}\dot{x}_k^s \\ \dot{y}_k^s \\ \dot{\psi}_k^s\end{bmatrix} = \begin{bmatrix}\frac{1}{\Delta t}\left(x_k^s - x_{k-1}^s\right) \\ \frac{1}{\Delta t}\left(y_k^s - y_{k-1}^s\right) \\ {}^{\{s\}}\dot{\psi}_k^s\end{bmatrix} + \begin{bmatrix}w_x^s \\ w_y^s \\ {}^{g}v^s\end{bmatrix}, \tag{17}$$

where ${}^{\{G\}}_{\{U\}}\mathbf{R}$ is a rotation from the ROV frame to the global frame, assuming that the INS (ins) is oriented identically to the ROV frame. Measurements from the gyroscope on each vehicle and the INS on the ROV are used as the "controls" for their angular velocities and accelerations respectively. The other sensors on each vehicle are used in the correction step, where their sensor models are given as:

$$\begin{aligned}
{}^{gps}\mathbf{z}^s\left(\mathbf{x}_k^s, {}^{gps}\mathbf{v}_k^s\right) &= \begin{bmatrix}1 & 0 & 0 \\ 0 & 1 & 0\end{bmatrix}\mathbf{x}_k^s + {}^{gps}\mathbf{v}_k^s \\
{}^{co}\mathbf{z}^s\left(\mathbf{x}_k^s, {}^{co}\mathbf{v}_k^s\right) &= \begin{bmatrix}0 & 0 & 1\end{bmatrix}\mathbf{x}_k^s + {}^{co}\mathbf{v}_k^s \\
{}^{co}\mathbf{z}^u\left(\mathbf{x}_{1,k}^u, {}^{co}\mathbf{v}_k^u\right) &= \begin{bmatrix}0 & 1 & 0 & 0\end{bmatrix}\mathbf{x}_{1,k}^u + {}^{co}\mathbf{v}_k^u \\
{}^{pr}\mathbf{z}^u\left(\mathbf{x}_{1,k}^u, {}^{pr}\mathbf{v}_k^u\right) &= \begin{bmatrix}1 & 0 & 0 & 0\end{bmatrix}\mathbf{x}_{1,k}^u + {}^{pr}\mathbf{v}_k^u.
\end{aligned}$$

where gps, co and pr represent the GPS, compass and pressure sensors respectively.

B. Grid Based Estimation of ROV Position

The motion model of the ROV in the x, y plane, $\dot{\mathbf{x}}_{2,k}^u$, is given as

$$\mathbf{V} = \begin{bmatrix}0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1\end{bmatrix}$$
$$\begin{bmatrix}\dot{x}_k^u \\ \dot{y}_k^u\end{bmatrix} = \mathcal{N}\left(\mathbf{V}\mathbf{x}_{1,k|k}^u, \mathbf{V}\mathbf{\Sigma}_{k|k}^u\mathbf{V}^\top\right),$$

where the right hand side is the velocity estimated from the EKF in the previous step. $\mathbf{V}$ is a matrix used to extract just the velocity estimate. Note that the velocity is assumed to be independent of vertical position and yaw over the small timestep of estimation given the arrangement of thrusters on the ROV.

The camera and tether are both sources that make the estimate of the ROV's position non-Gaussian. In order to introduce the tether in the generalized framework (7), the tether model is defined as

$$\begin{aligned}
&{}^{t}l_k(\mathbf{x}_k^u|\mathbf{x}_k^s, \mathbf{x}_{1,k}^u, {}^{t}\tilde{\mathbf{z}}_{1,k}^u) \\
&\qquad = \begin{cases}1 & \exists\, {}^{t}\tilde{\mathbf{z}}_{1,k}^u \in {}^{t}\mathcal{X}_d^u \\ 0 & \text{otherwise}\end{cases}
\end{aligned} \tag{18}$$

where the tether's "detectable region" is the circle defined by its length, the USV's position, and the ROV's depth at time k:

$${}^{t}\mathcal{X}_d^u = \left\{\hat{\mathbf{x}}_{i,k}^u \,\middle|\, ||\hat{\mathbf{x}}_{i,k}^u - \mathbf{x}_k^s|| < \sqrt{(L_k^t)^2 - (z_k^u)^2}\right\}. \tag{19}$$

The camera requires more work as it is able to provide different observations depending on whether or not it is able to detect the USV. Its model, that is described generically in Eq. (4), has the first equation as:

$$ {}^{c}\mathbf{z}^{s}\left(\mathbf{x}_k^s, \mathbf{x}_{2,k}^u, {}^{c}\mathbf{v}_k^u\right) = {}^{\{C\}}_{\{G\}}\mathbf{R}\left(\mathbf{x}_k^s - \mathbf{x}_{2,k}^u\right) - {}^{\{C\}}_{\{G\}}\mathbf{T} + {}^{c}\mathbf{v}_k^u \tag{20a} $$

where ${}^{\{2\}}_{\{1\}}\mathbf{T}$ is a translation from frame 1 to 2, and c is the camera. Finding the observable region ${}^{q_j}\mathcal{X}_o^r$ in Eq. (3) first requires defining the probability of detection for the camera. Based on prior work, this is given as $P_d(\mathbf{x}_k^s|\mathbf{x}_k^{u_c};\boldsymbol{\pi}^{u_c}) = exp(-1.25^{u_c}r^s)$ where ${}^{u_c}r^s$ is the range from the camera to a point on the surface, given by Eq. 21a. The detectable region ${}^{u_c}\mathcal{X}_d^s$ is then given by computing the probability of detection such that

$$ {}^{u_c}r^s = \frac{z^u}{\sin({}^{u_c}\zeta_p^s)\cos({}^{u_c}\theta_p^s)} \tag{21a} $$

$$ {}^{u_c}\zeta^s \in [{}^{u_c}\zeta_{min}^s, {}^{u_c}\zeta_{max}^s] \tag{21b} $$

$$ {}^{u_c}\theta^s \in [-{}^{u_c}\theta_{max}^s, {}^{u_c}\theta_{max}^s] \tag{21c} $$

where ${}^{u_c}\zeta^s$ is the azimuth angle from horizontal to a point and ${}^{u_c}\theta^s$ is the bearing reading from the camera to that same point. Note that this formulation assumes the camera is at a fixed angle on the front of ROV. To find the detectable region projected onto the x, y plane, the same constraints are applied to

$$ \begin{bmatrix} {}^{\{U\}u_c}x_p \\ {}^{\{U\}u_c}y_p \end{bmatrix} = \begin{bmatrix} {}^{u_c}r_p^s\cos({}^{u_c}\zeta_p^s)\cos({}^{u_c}\theta_p^s) \\ {}^{u_c}r_p^s\sin({}^{u_c}\theta_p^s) \end{bmatrix} + {}^{\{U\}}_{\{C\}}\mathbf{T} \tag{22a} $$

$$ {}^{c}\mathbf{x}_p = \mathbf{R}_{\pi}{}^{\{G\}}_{\{U\}}\mathbf{R}\begin{bmatrix} {}^{u_c}x_p \\ {}^{u_c}y_p \end{bmatrix} + {}^{\{G\}}_{\{S\}}\mathbf{T}, \tag{22b} $$

where $\mathbf{R}_\pi$ is a rotation of $180°$ which is combined with the transformation ${}^{\{G\}}_{\{S\}}\mathbf{T}$ to define the region of the x, y plane in which the ROV is able to detect the USV, and therefore gain information about its own position.

IV. Experiments in Lake Environment

To demonstrate the ability to incorporate the non-detection information from the ROV's camera, experiments were conducted in a lake environment with low visibility. The experiments were conducted using a BlueRobotics BlueROV2 ROV and a custom built USV as shown in figure 1. Two experiments were run with different paths for the ROV, one where it performed a typical lawnmower pattern and one where it explored a pier. Figure 4 shows the vehicles used as well as the lake where the experiments were conducted.

(a) Lake where testing was performed (b) Custom USV (c) BlueROV2 from BlueRobotics

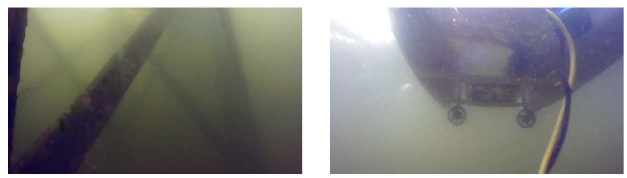
Fig. 4. (a) shows the lake where experiments were performed with the USV in (b); (c) shows the ROV that was used.

Table II shows the parameters that were used in the EKF and grid-based estimators for both runs.

TABLE II
Estimation parameters

	ROV	USV
Grid cell size	0.05m by 0.05m	
Tether length	5m	
$\boldsymbol{\Sigma}_k^{rw}$	$1\text{e}{-6}\text{m}^2$	$[1\text{m}^2, 1\text{m}^2]$
${}^{ins}\boldsymbol{\Sigma}_k^{uv}$	$[2\text{e}{-6}\frac{\text{m}^2}{\text{s}^4}, 2\text{e}{-6}\frac{\text{m}^2}{\text{s}^4}]$	N/A
${}^{g}\boldsymbol{\Sigma}_k^{rv}$	$1.5\text{e}{-6}$	$1.5\text{e}{-6}$
${}^{co}\boldsymbol{\Sigma}_k^{rv}$	$2.5\text{e}{-7}\text{rad}^2$	$2.5\text{e}{-7}\text{rad}^2$
${}^{gps}\boldsymbol{\Sigma}_k^{sv}$	N/A	$[0.56\text{m}^2, 0.56\text{m}^2]$
${}^{pr}\boldsymbol{\Sigma}_k^{uv}$	$6.5\text{e}{-6}\text{m}^2$	N/A

Figure 5 shows the view from the BlueROV2 in the lake where it is and is not able to detect the surface vehicle. An Apriltag was attached to the bottom of the USV to facilitate detection. Even in a shallow lake (maximum depth of approximately 5m), the ROV is only able to detect the USV for part of the area it is able to reach given the tether (it was typical able to detect the USV at up to 3.5m, while limited by the tether to 5m).



(a) Non-Detection View (b) Detection View

Fig. 5. Views from the BlueROV2 in the lake: (a) is a view where the USV is not in view, (b) shows a frame where the USV was detected on the camera.

For both runs, the ROV was also equipped with a Water Linked A50 Doppler Velocity Logger (DVL). The position estimate provided by this was used as the ground truth.

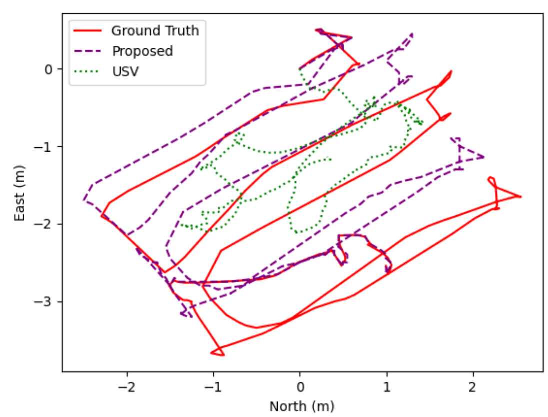


Fig. 6. Horizontal position estimate for the ROV during a lawnmower path.

Figures 6 and 7 show the path estimates for the ROV in the x, y plane alongside the ground truth and the USV's position. Despite only a handful of detections in each run (three in the lawnmower run and two while exploring the pier), the vehicle is able to maintain an absolute position error of less than $4\mathrm{m}$ in each run (below the constraint imposed by the tether).

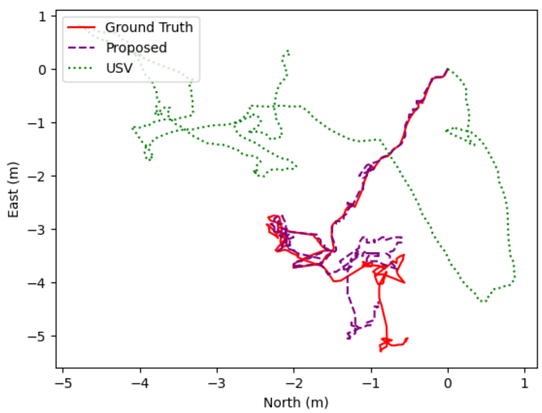


Fig. 7. Horizontal position estimate for the ROV while exploring a pier.

V. Conclusions

This paper has presented a technique for localizing an ROV cooperatively with a tethered USV by fusing the non-detection observations from the camera with the data from the ROV's INS in environments where there is limited visibility. It has described how the states are estimated using an EKF where observations are Gaussian and using a grid-based method where they are non-Gaussian, and specifically in the case of non-detection events of the camera. Validation in a pair of real world experiments showed the applicability of the approach.

Ongoing work on the approach includes Moving the velocity estimate into the grid-based method to estimate the dependence between position and velocity (for example, the tether constrains velocity at its edges), improving the model of the probability of detection based on turbidity and light level, and developing autonomous controls for the two vehicles that make use of the grid estimate.

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References

[1] C. Boittiaux, C. Dune, A. Arnaubec, R. Marxer, M. Ferrera, and V. Hugel, "Long-term visual localization in deep-sea underwater environment," in ORASIS. Carqueiranne, France: Thanh Phuong Nguyen, May 2023. [Online]. Available: https://hal.science/hal-04108737

[2] S. Rahman, A. Q. Li, and I. Rekleitis, "Svin2: An underwater slam system using sonar, visual, inertial, and depth sensor," in 2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, Nov. 2019, pp. 1861–1868.

[3] S. Hożyń and B. Żak, "Stereo vision system for vision-based control of inspection-class rovs," Remote Sensing, vol. 13, no. 24, p. 5075, Dec. 2021.

[4] H. Xing, L. Shi, K. Tang, S. Guo, X. Hou, Y. Liu, H. Liu, and Y. Hu, "Robust rgb-d camera and imu fusion-based cooperative and relative close-range localization for multiple turtle-inspired amphibious spherical robots," Journal of Bionic Engineering, vol. 16, no. 3, pp. 442–454, May 2019.

[5] Q. Wei, Y. Yang, X. Zhou, C. Fan, Q. Zheng, and Z. Hu, "Localization method for underwater robot swarms based on enhanced visual markers," Electronics, vol. 12, no. 23, p. 4882, Dec. 2023.

[6] M. Xanthidis, B. Joshi, J. M. O'Kane, and I. Rekleitis, "Multi-robot exploration of underwater structures," IFAC-PapersOnLine, vol. 55, no. 31, pp. 395–400, 2022.

[7] T. Furukawa, F. Bourgault, B. Lavis, and H. F. Durrant-Whyte, "Bayesian search-and-tracking using coordinated uavs for lost targets," in 2006 IEEE International Conference on Robotics and Automation, May 2006, pp. 1336–1341.

[8] T. Furukawa, L. C. Mak, H. F. Durrant-Whyte, and R. Madhevan, "Autonomous bayesian search and tracking and its experimental validation," Advanced Robotics, vol. 26, no. 5-6, pp. 461–485, 2012.